

Dynamic metrology (MO4006)

Solutions to the linear viscous Ekman spiral

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1 Theory of the Ekman spiral

We know that the wind can be written as

$$\mathbf{u} = u\hat{x} + v\hat{y} + w\hat{z}, \text{ where } u = u_0 + u'. \quad (1)$$

In the Ekman layer we assume that $w = 0$ and that the flow is steady, i.e. $\partial/\partial t = 0$. The equations of motion in a rotating frame of reference are then given by

$$\begin{cases} \frac{\partial^2 v}{\partial z^2} = \frac{f}{\nu} u, \\ \frac{\partial^2 u}{\partial z^2} = -\frac{f}{\nu} v. \end{cases} \quad (2)$$

with the boundary conditions

$$\begin{cases} u(z=0) = 0, \\ v(z=0) = 0, \\ u(z \rightarrow \infty) = u_g, \\ v(z \rightarrow \infty) = v_g. \end{cases} \quad (3)$$

Here, f is the Coriolis parameter and ν is the kinematic viscosity.

If the kinematic viscosity is assumed to be constant with height, the solution to these equations is the well-known Ekman spiral.

$$\begin{cases} u'(z) = -u_0 \exp\left(-\frac{z}{\delta}\right) \cos\left(\frac{z}{\delta}\right), \\ v(z) = u_0 \exp\left(-\frac{z}{\delta}\right) \sin\left(\frac{z}{\delta}\right), \end{cases} \quad (4)$$

where $\delta = \sqrt{2\nu/f}$ is the Ekman depth. However, in reality the viscosity varies with height.

2 Numerical solution using a collocation BVP solver

2.1 Problem

We solve the coupled ODE system on $z \in [0, \infty)$:

$$\begin{aligned} v''(z) &= a(z) u(z), \\ u''(z) &= -a(z) v(z), \end{aligned}$$

with boundary conditions at the surface

$$v(0) = 0, \quad u(0) = -u_0,$$

and the physical requirement

$$u(z) \rightarrow 0, \quad v(z) \rightarrow 0 \quad (z \rightarrow \infty).$$

2.2 The idea

1. Replace the infinite domain by a large finite height $z_{\max}$.
2. At $z_{\max}$ introduce a *decay condition* that goes to zero rather than forcing exact zeros there.
3. Convert the two second-order ODEs into a first-order system and solve the resulting boundary-value problem numerically (using a collocation BVP solver).

2.3 Why not force $u = v = 0$ at $z_{\max}$?

Forcing $u(z_{\max}) = v(z_{\max}) = 0$ at a finite depth often gives error in that the solution does not go to zero slowly, and the solver might increase afterwards (for example by oscillation).

2.4 Robin boundary condition

If a function $y(z)$ decays like $y \propto e^{-kz}$ for large z , then

$$y'(z) = -k y(z) \implies y'(z) + k y(z) = 0.$$

This is a first-order mixed Robin condition. For our system we know that we enter a region where $a = \text{constant}$, thus the solution becomes the classical Ekman spiral:

$$\begin{cases} v''(z) = a u(z), \\ u''(z) = -a v(z) \end{cases} \implies \begin{cases} u(z) = -u_0 \exp(-kz) \cos(kz), \\ v(z) = u_0 \exp(-kz) \sin(kz), \end{cases} \quad (5)$$

Where $k = \sqrt{a/2}$. Taking the derivative we obtain

$$u' = -k(u - v), \quad v' = -k(u + v).$$

Thus at $z = z_{\max}$ we impose the boundary conditions

$$u'(z_{\max}) + k u(z_{\max}) - k v(z_{\max}) = 0, \quad v'(z_{\max}) + k u(z_{\max}) + k v(z_{\max}) = 0.$$

Thus these new boundary conditions force the solver to choose solutions whose slope at high altitudes matches a decay rate k .

2.5 How to choose the decay rate

- If $a(z) = a_0$ is constant, then we get a decay rate of $k = \sqrt{a_0/2}$.
- If $a(z)$ varies, one must use the limit of k at the upper limit as $k = \sqrt{a_{z_{\max}}}/2$.

2.6 Rewrite as a first-order system

Define

$$Y(z) = \begin{bmatrix} u \\ u' \\ v \\ v' \end{bmatrix}. \quad (6)$$

Then the ODEs become

$$Y' = \begin{bmatrix} u' \\ -a(z)v \\ v' \\ a(z)u \end{bmatrix}. \quad (7)$$

We now have a first-order system $Y' = F(z, Y)$ and four boundary conditions.

$$\begin{cases} v(0) = 0, \\ u(0) = -u_0, \\ u'(z_{\max}) + k u(z_{\max}) - k v(z_{\max}) = 0, \\ v'(z_{\max}) + k u(z_{\max}) + k v(z_{\max}) = 0. \end{cases} \quad (8)$$

2.7 How the numerical solver (solve_bvp) works

Newton's method (or a Newton-like iteration) is used to solve that algebraic system. This requires an initial guess for the full solution profile. We use

$$u \sim -u_0 e^{-kz} \cos(kz), \quad v \sim u_0 e^{-kz} \sin(kz). \quad (9)$$

the solver returns a polynomial solution.

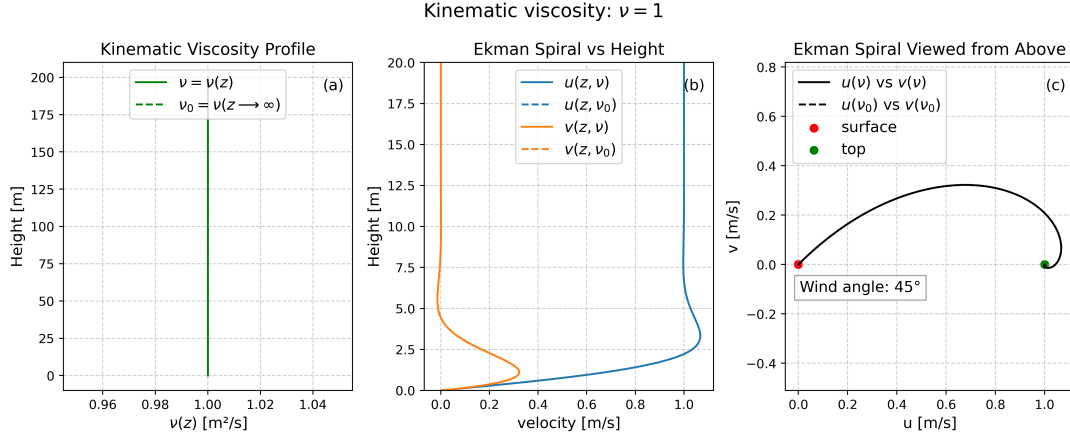


Figure 1: This figure shows the numeric solution for a constant profile.

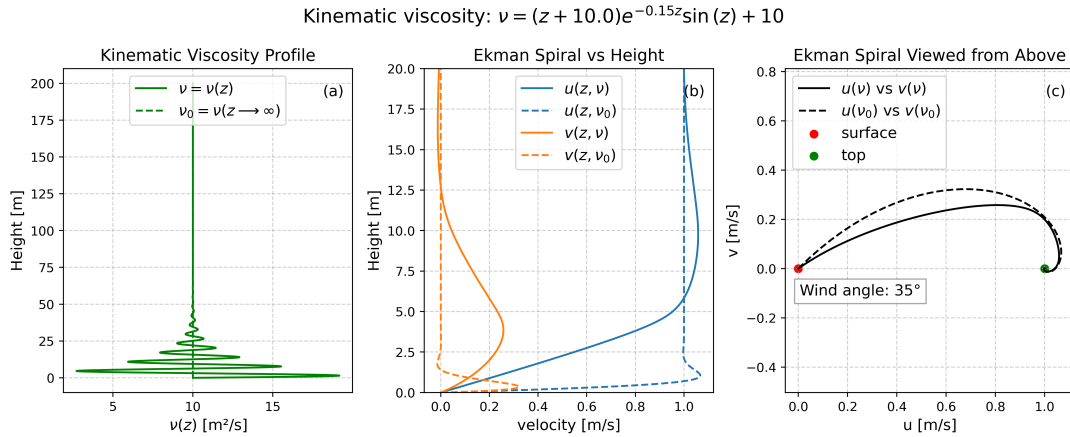


Figure 2: This figure shows the numeric solution for another specific profile.

3 logarithmic wind law

From the log-wind law we know

$$\mathbf{u} = \frac{u_*}{k} \ln\left(\frac{z}{z_0}\right) \Rightarrow \frac{\partial \mathbf{u}}{\partial z} = \frac{u_*}{kz} \quad (10)$$

However we also know from Prandtl that

$$K_m = l^2 \left| \frac{\partial \mathbf{u}}{\partial z} \right| \quad (11)$$

using these two relations we get together with the assumption that $l = kz$ we obtain

$$K_m = (kz)^2 \frac{u_*}{kz} = ku_* z \quad (12)$$

This K_m is the turbulent viscosity in the boundary layer. However, we know that this should start to decay after a certain height, and one can thus assume decay so that the total kinematic viscosity is zero at height $z = H$ as:

$$\nu = \kappa u_* z \left(1 - \frac{z}{H}\right) \quad (13)$$

or one could expect a exponential decay at as

$$\nu = \kappa u_* z e^{z/H} \quad (14)$$

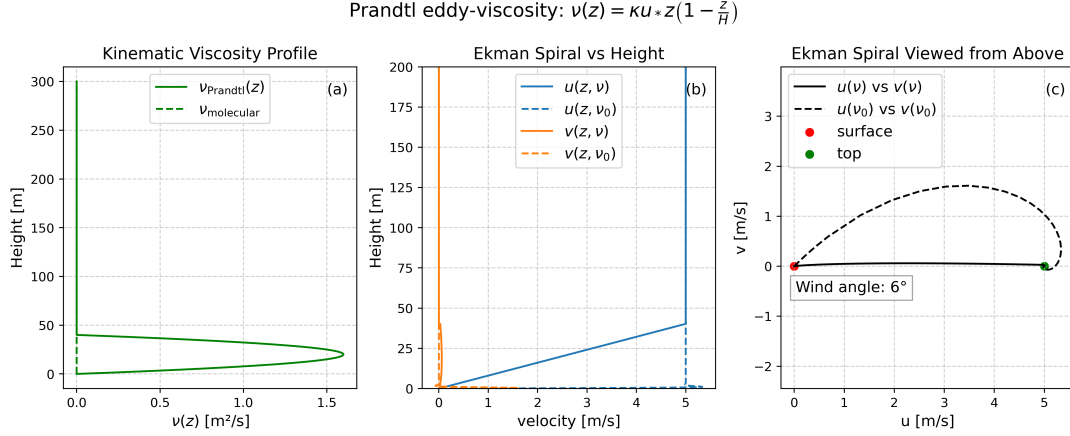


Figure 3: This figure shows the numeric solution for another specific profile.

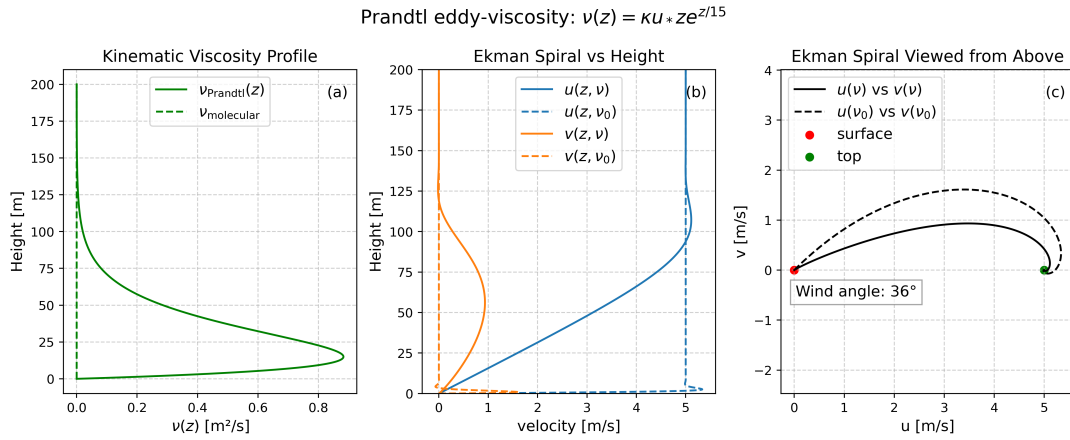


Figure 4: This figure shows the numeric solution for another, more realistic, profile.